1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlighted that some NMIBC tumors with favourable local immune context (high Immunoscore/TILs) nonetheless fail to obtain durable benefit from immune-based therapies — why does pre-existing immune infiltration not always translate into therapy sensitivity? 2) Tier 1 paths used: I built directly on the Immunoscore (operationalized here as TIL density) because it provides the imaging readout of local adaptive immunity, and on MSH6 (MMR component) because it mechanistically modulates neoantigen burden and therefore the quality of anti-tumor T-cell responses. These two form a parsimonious mechanistic axis: immune presence (TILs) × neoantigen generation (MSH6 status) → productive anti-tumor immunity. 3) Tier 2 evidence that supports and refines the claim: ADNP mRNA in CTCs and stool methylation markers (SDC2/TFPI2) provide non-invasive, longitudinal measures of tumor burden/biology that could validate dynamic changes in immune engagement; HDL-C and AI-enabled MSI readouts offer systemic and computational layers that can modulate or operationalize the Immunoscore–MSH6 interaction in real-world deployment. 4) Why I chose therapy_response as the endpoint: the central translational question from the Ontologist concerns prediction of immunotherapy benefit and treatment selection in NMIBC — a therapy_response proxy is the most actionable endpoint to test the mechanistic interaction between TILs and MSH6. 5) What I excluded and why: I deliberately excluded exploratory histopathologic patterns (e.g., SARIFA), hormone-pathway/TP53 links, and large multi-marker metabolic panels from the core hypothesis to keep it mechanistically sharp and testable; these remain as secondary_modifiers or future directions rather than part of the 1-imaging + ≤2-anchor gene core?

Working Hypothesis

In NMIBC, baseline TIL density predicts immunotherapy response only when tumors harbor defective MSH6-mediated DNA mismatch repair, which elevates neoantigen burden and promotes productive T-cell activity. Conversely, NMIBC with high TIL density but intact MSH6 will show diminished immunotherapy benefit due to lower neoantigen load and potential T-cell exhaustion. Therefore, the predictive value of TIL density is determined by an interaction with MSH6 status, such that integrating TIL density with MSH6 status provides a more accurate prediction of immunotherapy response than either biomarker alone. Secondary modifiers (e.g., ADNP expression and DNA methylation signatures) may refine this prediction but are not required for the core hypothesis.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 (MMR component) because it mechanistically modulates neoantigen burden and therefore the quality of anti-tumor T-cell responses.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	ADNP (CTCs)	Supporting modifier from knowledge graph
Secondary	SDC2 methylation	Supporting modifier from knowledge graph
Secondary	TFPI2 methylation	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	AI-based MSI biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- ADNP (CTCs): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SDC2 methylation: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.